Diffusion Generative Models

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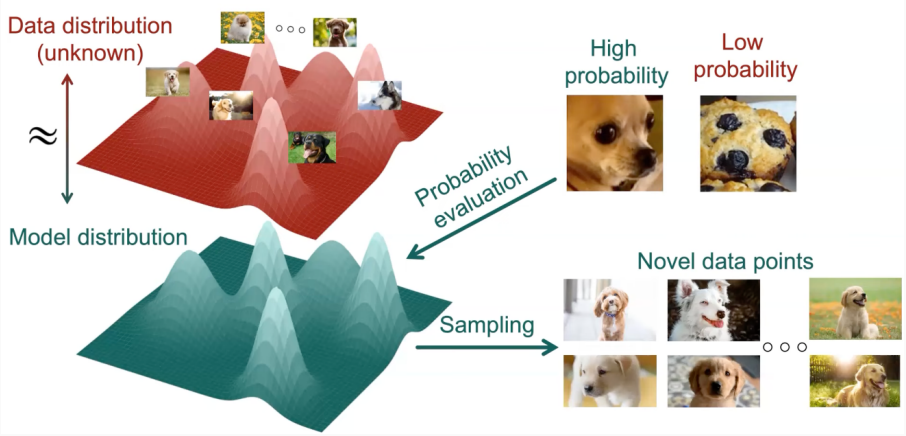
Introduction and Motivation

- Understand the weaknesses of existing generative models
- Learn about diffusion models and their SOTA performance
- Explore fundamentals of score matching and SDEs
- Gain practical insights into implementing diffusion models

What is Generative Modelling?

Given a dataset $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$ of N samples $\mathbf{x}_i \in \mathbb{R}^d$:

1. Approximate a latent probability density function (PDF) $p_{\text{data}}(x)$ that describes the data.
2. Generate novel samples similar to those in $\mathcal{D}$.
3. Identify latent generative factors in the data.



A latent PDF of dog images [8].

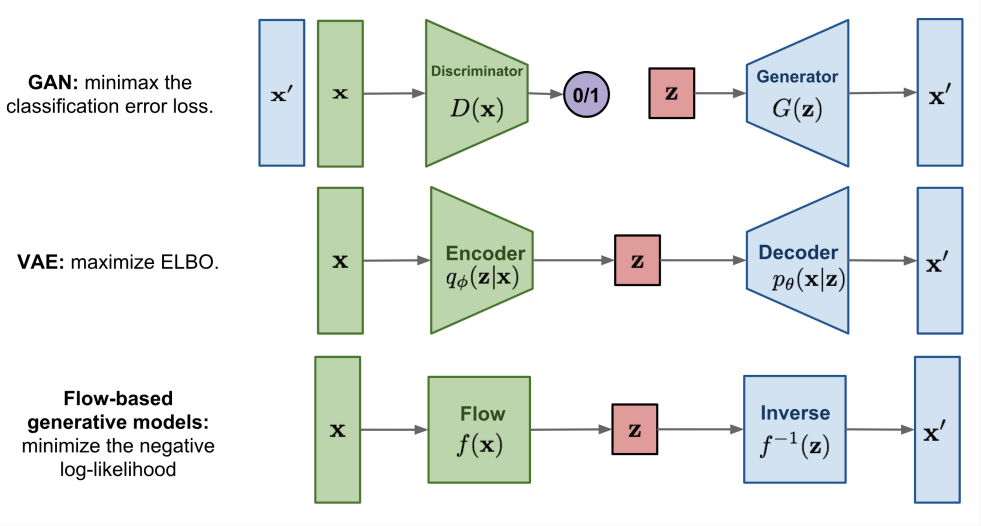
Constraints of Prior Generative Models

1. Generative Adversarial Networks (GANs) are prone to mode collapse.
2. Variational Autoencoders (VAEs) require a *normalising constant* and a *surrogate objective*.
3. Flow-based models require a normalising constant and *bijectivity*.

Def. (Normalising Constant)

$$Z(\theta) := \int_{\mathbb{R}^n} f(\theta, \mathbf{x}) \mathrm{d}\mathbf{x},$$

s.t. $f : \mathbf{\Theta} \times \mathbb{R}^n \to \mathbb{R}$ and $p(\theta, \mathbf{x}) := \frac{f(\theta, \mathbf{x})}{Z(\theta)}$ is a PDF.



Comparing generative models [9].

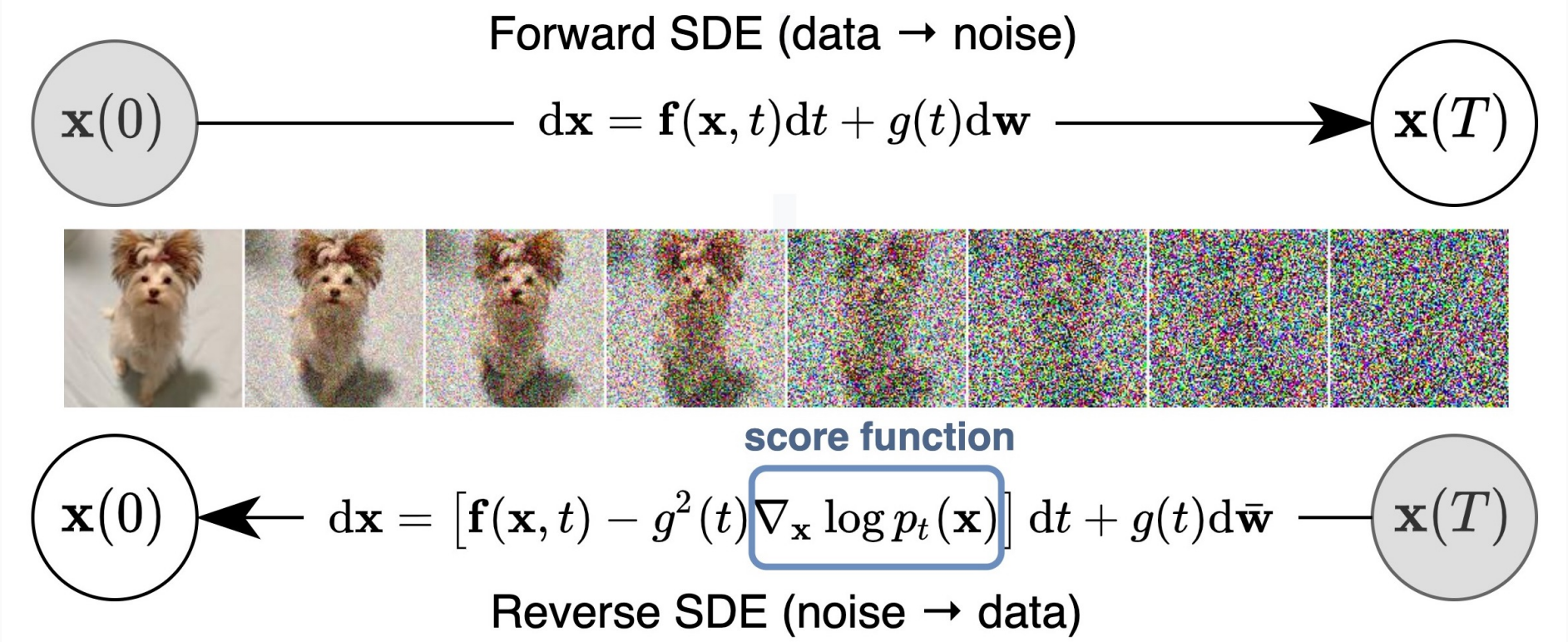
Introduction to Diffusion Models

1. Turns noise into data by learning the *score*.
2. Learns the score by turning data into noise.
3. No need for normalising constant or bijectivity.
4. Relies on stochastic differential equations (SDEs).
5. SOTA performance in a variety of tasks.



A food dye analogy: training from left to right, generative modelling from right to left.

Diffusion Schematic



Data turns into a Gaussian by forward diffusion. A sample is taken from the Gaussian and turned into similar data by reverse diffusion [10].

The Score and Normalising Constant

Def (Score function)

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \approx s(\theta, \mathbf{x}, t)$$

where $p(\mathbf{x}, t)$ is the marginal PDF of $\mathbf{x}(t)$ evolving over time $t \in \mathbb{R}$ with the solution of the SDE.

Task

Show that the score doesn't depend on the normalising constant:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log \frac{f(\mathbf{x}, t)}{Z}$$

$$\dots$$

Forward and Reverse Diffusion Processes

Def. (Forward Diffusion)

A stochastic differential equation (SDE) for $\mathbf{x}(t) \in \mathbb{R}^n$ with $t \in \mathbb{R}$

$$\mathrm{d}\mathbf{x} = \mathbf{F}(\mathbf{x}, t)\mathrm{d}t + \mathbf{G}(\mathbf{x}, t)\mathrm{d}\mathbf{w}$$

where $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ and $\mathbf{G} : \mathbb{R}^n \to \mathbb{R}^{n \times n}$ are Lipschitz-continuous coefficients, and $\mathbf{w}$ is a standard Wiener process.

Theorem (Reverse Diffusion [1, 4, 10])

The reversal of the forward diffusion for $\mathbf{x}(\tau) \in \mathbb{R}^n$ with $\tau := T - t$ with $T \in \mathbb{R}$

$$\mathrm{d}\mathbf{x} = \left[-\mathbf{F}(\mathbf{x}, \tau) + \nabla_{\mathbf{x}} \cdot \mathbf{G}(\mathbf{x}, \tau)\mathbf{G}(\mathbf{x}, \tau)^\top + \mathbf{G}(\mathbf{x}, \tau)\mathbf{G}(\mathbf{x}, \tau)^\top \nabla_{\mathbf{x}} \log p(\mathbf{x}, \tau)\right] \mathrm{d}\tau + \mathbf{G}(\mathbf{x}, \tau)\mathrm{d}\mathbf{w}$$

where the marginal PDF $p(\mathbf{x}, t)$ is equivalent for the forward diffusion and we wish to learn $s(\theta, \mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$.

Training with Score Matching

Algorithm

1. Generate *forward diffusion* trajectories $\mathbf{x}(t)$ such that $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$ and $\mathbf{x}(T) \sim p_{\text{model}}(\mathbf{x})$.
2. Train a score model $\mathbf{s}(\theta, \mathbf{x}, t)$ on the forward trajectories until $s(\theta, \mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$.
3. Generate novel data by generating *reverse diffusion* trajectories $\mathbf{x}(\tau)$ such that $\mathbf{x}(T) \sim p_{\text{model}}(\mathbf{x})$ and $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$.

Questions

1. How do we train the score model $\mathbf{s}(\mathbf{x}, t)$ when we don't know $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$?
2. How do we ensure that $p_{\text{model}}(\mathbf{x})$ is tractable for initialising the reverse diffusion?

How to Train the Score Model?

Def. (Explicit Score Matching [5])

$$\mathcal{L}_{\text{ESM}}(\theta) = \frac{1}{2}\mathbb{E}_{t\sim\mathcal{U}(0,1)}\left[\|\mathbf{s}(\theta,\mathbf{x}(t),t) - \nabla_{\mathbf{x}}\log p(\mathbf{x}(t),t)\|^2\right],$$

where $t \sim \mathcal{U}(0,1)$ is a random time uniformly distributed between 0 and 1 and $\mathbf{x}(t)$ are trajectories sampled from the forward diffusion.

Theorem (Implicit Score Matching [5])

$$\mathcal{L}_{\text{ISM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,1)}\left[\text{Tr}\left[\nabla_{\mathbf{x}}\mathbf{s}(\theta,\mathbf{x}(t),t)\right] + \frac{1}{2}\|\mathbf{s}(\theta,\mathbf{x}(t),t)\|^2\right] + C,$$

where C is a constant.

Task

Prove the theorem. See [5] for hints.

How to Train the Score Model?

Remark

$\mathrm{Tr}\left[\nabla_{\mathbf{x}}\mathbf{s}(\theta, \mathbf{x}(t), t)\right]$ is expensive to compute.

Theorem (Sliced Score Matching [11])

$$\mathcal{L}_{\mathrm{SSM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,1),\mathbf{v}\sim\mathcal{N}(\mathbf{0},\mathbf{I})}\left[\mathbf{v}^\top \nabla_{\mathbf{x}}\mathbf{s}(\theta, \mathbf{x}(t), t) + \frac{1}{2}\left(\mathbf{v}^\top \mathbf{s}(\theta, \mathbf{x}(t), t)\right)^2\right],$$

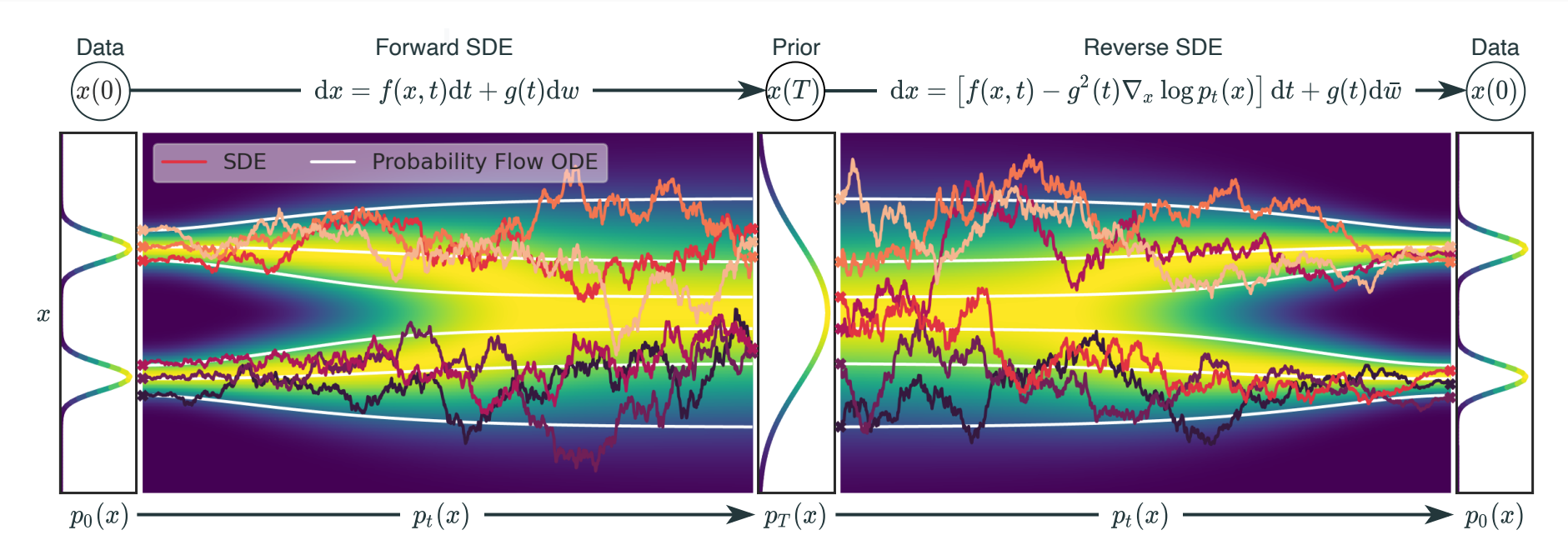
where $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a random direction. Good when the diffusion is *nonlinear*.

Theorem (Denoising Score Matching [13])

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(0,1)}\left[\left\|\mathbf{s}(\theta, \mathbf{x}(t), t) - \nabla_{\mathbf{x}} \log p(\mathbf{x}(t) \mid \mathbf{x}(0))\right\|^2\right],$$

where $p(\mathbf{x}(t) \mid \mathbf{x}(0), t)$ is analytical if the forward diffusion is *linear*.

Linear Diffusion



The data distribution gets transformed into a tractable Gaussian.

How to ensure $p_{\text{model}}(\mathbf{x})$ is tractable?

Def. (Linear Forward Diffusion)

$$\mathrm{d}\mathbf{x} = \mathbf{F}(t)\mathbf{x}\mathrm{d}t + \mathbf{G}(t)\mathrm{d}\mathbf{w}$$

Theorem (Linear Transition Density [7])

A normal distribution

$$p(\mathbf{x}(t) \mid x(0)) = \mathcal{N}\left(\mu(t), \mathbf{\Sigma}(t)\right),$$

where $\mu(t)$ and $\mathbf{\Sigma}(t)$ are solutions to

$$\frac{d\mu}{dt} = \mathbf{F}(t)\mu(t), \quad \frac{d\mathbf{\Sigma}}{dt} = \mathbf{F}(t)\mathbf{\Sigma}(t) + \mathbf{\Sigma}(t)\mathbf{F}(t)^\top + \mathbf{G}(t)\mathbf{G}(t)^\top,$$

with $\mu(0) \sim p_{\text{data}}$ and typically $\mathbf{\Sigma}(0) = 0$.

Example: Computing Mean and Covariance for a Given SDE

Example

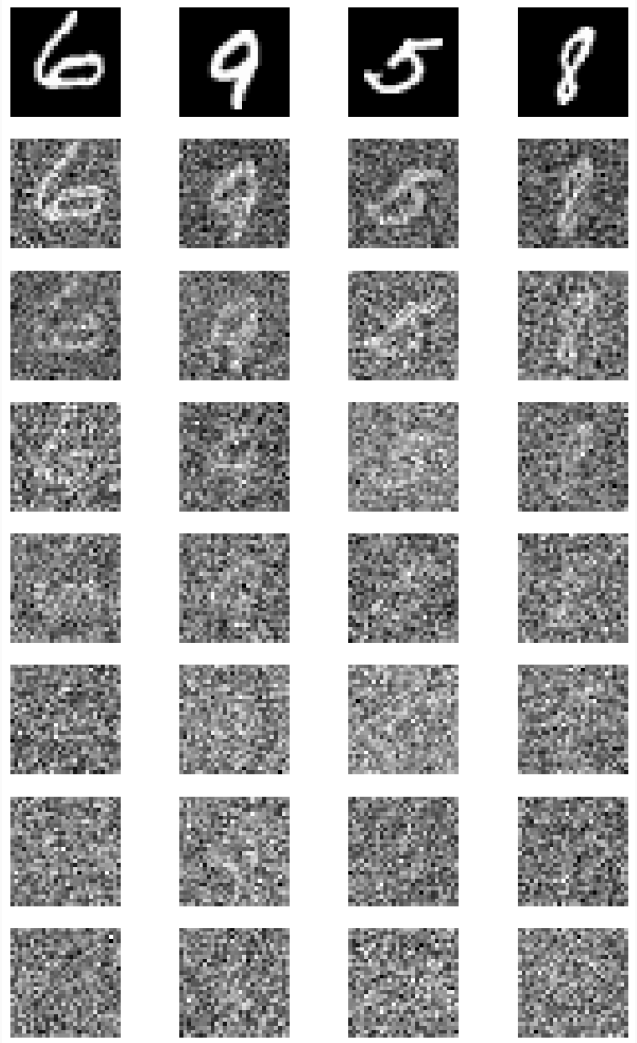
A linear forward diffusion with $\mathbf{F}(t) = \mathbf{0}$ and $\mathbf{G}(t) = \sigma^{2t}\mathbf{I}$:

$$\mathrm{d}\mathbf{x}(t) = \sigma^{2t}\mathrm{d}\mathbf{w} \quad \text{s.t.} \quad \sigma \in \mathbb{R}_{>0}.$$

Task

Assuming $\mu(0) \sim p_{\text{data}}$ and $\mathbf{\Sigma}(0) = \mathbf{0}$, show that

$$\mu(T) \approx \mathbf{0}, \quad \mathbf{\Sigma}(T) = \frac{\sigma^{4T} - 1}{4 \log \sigma}\mathbf{I}.$$



Adding noise to MNIST digits with $\sigma = 2$.
Eventually, $x(T) \sim p_{\text{model}} = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}(T))$.

Generating Data Samples

Remark

We generate samples by simulating the reverse diffusion from $\mathbf{x}(T) \sim p_{\text{model}}$ to $x(0)$, using the following methods.

Def. (Euler-Maruyama Method)

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{F}(\mathbf{x}_n, t_n)\Delta t + \mathbf{G}(\mathbf{x}_n, t_n)\Delta \mathbf{W}_n$$

where $\Delta t \in \mathbb{R}$ is a time step and $\Delta \mathbf{W}_n \sim \mathcal{N}(0, \Delta t \mathbf{I})$ is a Wiener increment.

Theorem (Probability Flow ODE [10])

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t) - \frac{1}{2}\nabla \cdot (\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^\top) - \frac{1}{2}\mathbf{G}(\mathbf{x}, t)\mathbf{G}(\mathbf{x}, t)^\top \nabla \log p(\mathbf{x}, t),$$

which can be simulated with any black-box ODE integrator (e.g. RKF45).

Conclusions

Key points

1. Diffusion models sample from the data distribution by turning noise into data.
2. Linear diffusion is the most practical and popular.
3. Easier to train than GANs and flow-based models.
4. No need for bijectivity or normalising constant.
5. SOTA generative model (e.g. beats GANs on CIFAR-10).

More info

1. SOTA performance in e.g. biology [2] and art [6].
2. More complicated diffusion can be used, e.g. Hamiltonian-based [3].
3. Can be used within latent space [12].

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